Uncertainty Quantification of Electric Load Demand Forecasts Using Probabilistic Deep Learning Models

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***Abstract*—Accurate and reliable load forecasting is critical for power system planning, operations, and real-time decision-making. However, point forecasts fail to capture the uncertainty inherent in load forecasts under variable conditions. To address this limitation, this paper presents a comparative analysis of deterministic and probabilistic Long Short-Term Memory (LSTM) based approaches for quantifying uncertainty in short-term load forecasting. The forecasting models are built upon the LSTM neural network architecture and trained on hourly electric load and temperature data. A baseline deterministic LSTM model is first developed to optimize point forecasting accuracy. Building on this foundation, two probabilistic variants are developed: (i) a parametric LSTM model with Gaussian outputs, trained using the negative log-likelihood (NLL) loss function to directly learn predictive distributions, and (ii) a Monte Carlo (MC) dropout-based LSTM, which estimates uncertainty through repeated stochastic forward passes during inference. The models are rigorously evaluated in terms of both deterministic forecast accuracy and the quality of the generated prediction intervals (PIs), measured by coverage probability and interval sharpness across multiple forecast horizons. Experimental results demonstrate that both probabilistic LSTM models effectively capture uncertainty while maintaining competitive deterministic forecasting performance relative to the deterministic model. These findings underscore the potential of deep probabilistic forecasting to enhance situational awareness and risk-informed decision-making in modern power systems.**

***Index Terms*—Deep learning model, electric load forecasting, long short-term memory (LSTM), probabilistic forecasting, uncertainty quantification.**

I. Introduction

Accurate electric load forecasting plays a pivotal role in power system operations, directly influencing generation scheduling, unit commitment, and real-time dispatch [1]. As modern power grids transition toward greater integration of variable renewable energy sources and increasingly dynamic consumption patterns, the need for reliable and robust forecasting methodologies becomes more critical than ever. Renewable generation technologies such as wind and solar inherently introduce generation fluctuations, while the growing penetration of electric vehicles and smart appliances adds significant variability and complexity to demand-side behavior [2]. Historically, a wide range of classical time series forecasting models such as autoregressive integrated moving average (ARIMA), exponential smoothing, and support vector regression (SVR) have been applied in electric load forecasting due to their simplicity and interpretability [3], [4]. However, these models often struggle to capture the nonlinearities and long-term temporal dependencies present in today's high-resolution, data-rich power systems. In recent years, machine learning (ML) and, in particular, deep learning (DL) techniques have demonstrated superior performance in forecasting applications by effectively modeling complex, high-dimensional time-series data [5], [6].

Among DL architectures, Long Short-Term Memory (LSTM) networks have gained widespread use for load forecasting tasks due to their ability to retain temporal context and mitigate vanishing gradient issues via gated recurrent units [7], [8]. Despite their predictive strength, most LSTM-based approaches remain deterministic, producing single-point forecasts without quantifying the inherent uncertainty. This limitation is especially problematic during periods of peak demand or under stochastic operating conditions, where deterministic forecasts alone may lead to suboptimal or risky decisions. To address this gap, probabilistic load forecasting has emerged as a critical extension, enabling the generation of predictive distributions rather than isolated deterministic estimates. This approach provides decision-makers with valuable uncertainty information, such as prediction intervals (PIs), which can support applications like reserve margin planning, demand response, and market bidding. Hong *et al.* [9] and Lauret *et al.* [10] proposed various methods in this domain, including quantile regression, Bayesian neural networks, and ensemble-based learning frameworks.

More recently, probabilistic deep learning models have gained traction for their capacity to model both types of uncertainty, i.e., randomness in the data and uncertainty due to limited knowledge in the model. Wen *et al.* [11] evaluated uncertainty quantification in solar PV forecasting using bootstrap and quantile regression methods with a wavelet-transformed, PSO-optimized RBFNN model, highlighting the growing need for more advanced deep learning-based sequence models that can directly model both data variability and uncertainty in forecasting. In this work, we incorporate two such approaches within an LSTM architecture. The first is a Gaussian-output LSTM trained with negative log-likelihood (NLL) loss to

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capture aleatoric uncertainty by parameterizing a distribution over the forecast outputs. The second approach enhances this with Monte Carlo (MC) dropout, which approximates Bayesian inference through stochastic forward passes during inference, enabling estimation of epistemic uncertainty [12], [13]. Together, these methods provide a basis for comparing deterministic and probabilistic LSTM approaches for electric load forecasting to assess both deterministic accuracy and uncertainty quantification.

This paper presents a comparative analysis of deterministic and probabilistic LSTM models for short-term (day-ahead and week-ahead) electric load forecasting and associated uncertainty quantification. The major contributions of this paper lie in the following: (i) Contribution to the field of power system forecasting problems by addressing uncertainty in evolving load demand patterns and supporting real-time planning and operational decision-making through the application of deterministic and probabilistic LSTM models to electric load demand forecasting; (ii) Development of probabilistic deep learning models to generate not only deterministic forecasts but also associated confidence intervals for uncertainty-aware predictions, thereby contributing to the field of predictive data analytics; and (iii) Systematic evaluation of three LSTM models (i.e., deterministic, Gaussian probabilistic, and MC Gaussian probabilistic) under diverse seasonal, holiday, and demand-extreme scenarios, highlighting the deep learning models' effectiveness in real-world forecasting conditions.

The rest of the paper is organized as follows. Section II presents the proposed forecasting methodology including data preprocessing and model design. Section III describes the evaluation metrics and experimental setup. Forecasting results and comparative analysis are discussed in Section IV. Section V concludes with key findings and future research directions.

II. Proposed Approach for Load Forecasting

This section details the methodology employed in this paper for forecasting load as illustrated in Fig. 1, which outlines the sequence of steps beginning from raw data ingestion to model training and evaluation.

The proposed forecasting framework begins with a year-long dataset that contains hourly electric load and temperature measurements, which serve as temporal and contextual features for model training. The raw data is processed and transformed through a structured preprocessing pipeline to prepare it for learning, including normalization, supervised sequence construction, and careful partitioning into training, validation, and test sets. Following data preparation, three LSTM-based models are developed to capture both deterministic and probabilistic aspects of load forecasting. The first is a standard LSTM trained for deterministic prediction, serving as a baseline model. To account for uncertainty, a Gaussian-output LSTM is introduced, which learns a predictive distribution over future load values. To further incorporate uncertainty, the third model extends the Gaussian framework by integrating Monte Carlo dropout during inference, allowing uncertainty quantification through stochastic ensembling.

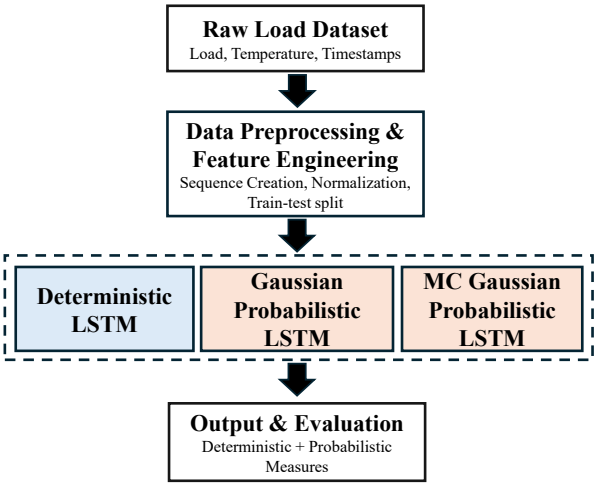

Fig. 1. Workflow of the proposed load forecasting framework.

Finally, the performance of all models is assessed using a combination of deterministic and probabilistic evaluation metrics. Deterministic accuracy is measured using Normalized Root Mean Squared Error (NRMSE) and Mean Absolute Percentage Error (MAPE), while probabilistic performance is evaluated through Prediction Interval Coverage Probability (PICP) and Continuous Ranked Probability Score (CRPS). This comprehensive approach enables a rigorous comparison between conventional and uncertainty-aware deep learning models for electric load forecasting.

A. Data Preprocessing and Feature Engineering

The dataset used in this paper consists of hourly measurements of electric load (in MW) and ambient temperature (in Fahrenheit) for the year 2011. Data was acquired from the ISO−New England electricity market [14]. Forecasting is performed for two target horizons, i.e., 24-hour (day-ahead) and 168-hour (week-ahead) forecasts. All continuous features are scaled to the [0, 1] range using min-max normalization based on the statistics of the training set to ensure stable and efficient model convergence, which is defined as

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{1}$$

where x is the original value of data; $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of the features, respectively; and x' is the normalized value scaled to the range $[0, 1]$.

Supervised learning sequences are generated using a sliding window approach, where each input sample consists of the past 168 hours of hourly electric load and temperature data. This input window is used to predict either a 24-hour-ahead or 168-hour-ahead sequence of future load values, depending on the forecasting horizon. To evaluate the generalization performance of the models under diverse operating conditions, representative test periods are selected in advance. These include one typical day and week from each season, the highest and lowest demand day and week of the year, and a major holiday day and week. These selected test periods are

kept separate for evaluation purposes. For each test window, the model is trained using the preceding month of historical data, providing sufficient temporal context while avoiding leakage from future data. This strategy ensures consistent and meaningful performance evaluation across different forecasting scenarios.

B. Deterministic Forecasting Using LSTM Model

The first model developed in this paper is an LSTM network used as a baseline for deterministic forecasting. It takes as input a sequence of 168 hourly values, including both electric load and temperature, and produces a fixed-length output corresponding to the forecasting horizon for a day-ahead or week-ahead forecast.

The LSTM network consists of an LSTM layer with 64 hidden units, followed by a dropout layer with a rate of 0.1 to mitigate overfitting. The output is generated through a fully connected dense layer whose number of neurons matches the desired forecast length. The model is trained using the mean squared error (MSE) loss function, optimized via the Adam optimizer, and defined as

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2}$$

where y_i is the actual load, $\hat{y}_i$ is the forecasted load, n is the number of forecasts, and $\bar{y}$ is the mean of the actual values. Training is performed over a maximum of 100 epochs with a batch size of 64 that considered an early stopping criterion using 15 epochs to prevent overfitting and retain the best-performing weights based on validation loss [15].

C. Probabilistic Forecasting Using Gaussian LSTM

To capture uncertainty inherent in electric load data, the deterministic LSTM architecture is reformulated to output parameters of a Gaussian distribution at each forecast timestep. Specifically, the model is modified to predict both the mean (μ) and the logarithm of the standard deviation ($\log\sigma$), thereby enabling it to learn a time-dependent distribution over future load values rather than a single point estimate [16]. The final dense layer in this architecture is designed to output $2\times$output_steps values—one mean and one log standard deviation for each forecasted time step. For instance, in the case of a 168-hour week-ahead forecast, the output dimension is 336. The model is trained using the negative log-likelihood (NLL) loss function, which is well-suited for learning Gaussian distributions. The NLL loss is defined as [17]

$$\mathcal{L}_{\text{NLL}} = \sum_t \left[\frac{1}{2}\log\sigma_t^2 + \frac{(y_t-\mu_t)^2}{2\sigma_t^2} + \frac{1}{2}\log(2\pi)\right] \tag{3}$$

where $\mathcal{L}_{\text{NLL}}$ is the negative log-likelihood loss, y_t is the observed load value at time t, μ_t is the forecasted load demand mean value, and σ_t is the predicted variance of the Gaussian distribution. The first term in the loss penalizes large predicted variances, encouraging the model to remain confident when possible. The second term acts as a scaled error penalty, weighting prediction errors inversely by the predicted variance. This formulation allows the model to express its confidence dynamically and is particularly useful for operational scenarios where understanding the range of possible future load values is critical.

D. Probabilistic Forecasting Using MC Gaussian LSTM

Probabilistic variability, which stems from model uncertainty and limited training data, is addressed by extending the Gaussian LSTM with MC dropout during inference. This approach enables approximate Bayesian inference by activating dropout layers during prediction and performing multiple stochastic forward passes. Each pass yields slightly different predictions due to random dropout masks, allowing the model to estimate variability across forecasts. Let N denote the number of forward passes performed during inference. Uncertainty is estimated as the variance of these predicted means and computed as the average of the predicted variances across all passes. The total predictive uncertainty is then calculated as the sum of these two components. This process is expressed mathematically as follows

$$\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N}\mu_i \tag{4}$$

$$\hat{\sigma}^2_{\text{epistemic}} = \frac{1}{N}\sum_{i=1}^{N}(\mu_i - \hat{\mu})^2 \tag{5}$$

$$\hat{\sigma}^2_{\text{aleatoric}} = \frac{1}{N}\sum_{i=1}^{N}\sigma_i^2 \tag{6}$$

$$\hat{\sigma}^2_{\text{total}} = \hat{\sigma}^2_{\text{epistemic}} + \hat{\sigma}^2_{\text{aleatoric}} \tag{7}$$

where N denotes the number of stochastic forward passes, μ_i is the predicted mean from the i-th sample, σ_i^2 is the predicted variance from the i-th pass, $\hat{\mu}$ is the predictive mean, $\hat{\sigma}^2_{\text{epistemic}}$ represents epistemic uncertainty (randomness in prediction), $\hat{\sigma}^2_{\text{aleatoric}}$ represents aleatoric uncertainty (data noise), and $\hat{\sigma}^2_{\text{total}}$ is the total predictive uncertainty. The final predictive distribution remains Gaussian, now parameterized by the aggregated mean and the total uncertainty. This approach enables more reliable forecasting under uncertainty by accounting for both data variability and model confidence, making it particularly valuable for high-stakes operational decision-making in power systems.

E. Forecast Evaluation Metrics

To comprehensively assess the forecasting models, four widely used performance indices are employed: NRMSE, MAPE, CRPS, and PICP. The NRMSE is a normalized form of the Root Mean Squared Error and is computed as follows

$$\text{NRMSE} = \frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}}{\bar{y}} \times 100\% \tag{8}$$

It provides a scale-independent measure of error, allowing performance comparison across different seasons and load levels. MAPE quantifies the average magnitude of forecast error as a percentage, defined as

TABLE I
DAY-AHEAD FORECASTING PERFORMANCE ACROSS SEASONS

Model	Metric	Winter	Spring	Summer	Fall
LSTM	NRMSE (%)	2.24	4.04	2.61	5.32
	MAPE (%)	1.82	3.46	2.34	4.12
Gaussian LSTM	NRMSE (%)	3.02	3.26	4.70	6.57
	MAPE (%)	2.80	2.30	4.10	5.70
	CRPS (MW)	274.88	266.00	418.15	503.00
MC Gaussian LSTM	NRMSE (%)	3.20	3.60	4.60	6.10
	MAPE (%)	2.30	2.60	3.70	5.10
	CRPS (MW)	289.50	267.54	395.13	473.64

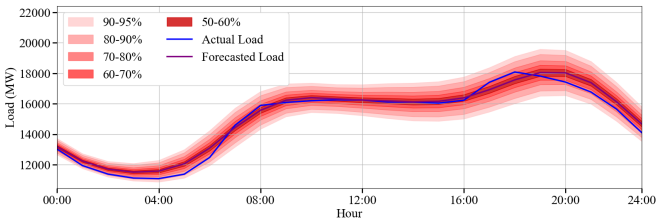

Fig. 2. Spring day-ahead forecasts using MC Gaussian LSTM.

$$\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right| \tag{9}$$

CRPS is used for evaluating the quality of probabilistic forecasts. It assesses the accuracy of the cumulative distribution function of the prediction compared to the actual outcome. A lower CRPS indicates a better calibrated and sharper predictive distribution. A formal definition and foundational treatment of CRPS is available in [18]. Additionally, we evaluate PICP, which measures the proportion of true values falling within the predicted prediction interval at a given confidence level. PICP offers insights into the reliability of the uncertainty bounds produced by probabilistic models and helps assess how well these intervals capture actual outcomes across different confidence levels [19]. These performance metrics provide complementary insights into the accuracy and reliability of both deterministic and probabilistic models. A broader overview and comparison of error metrics in energy forecasting can be found in [20].

III. FORECASTING RESULTS AND DISCUSSION

This section presents and analyzes the forecasting performance of three LSTM-based models i.e., deterministic LSTM, Gaussian LSTM, and MC Gaussian LSTM—under various practical scenarios. The evaluation is structured around four representative forecasting cases: day-ahead, week-ahead, holiday, and forecasting during extreme demand conditions. Each forecasting scenario is evaluated using both deterministic and probabilistic approaches. The deterministic LSTM generates point forecasts solely and is assessed using NRMSE and MAPE. In contrast, the Gaussian and MC Gaussian LSTM models produce probabilistic forecasts to allow the estimation of forecast distributions and these models are evaluated using PICP and CRPS which quantify the quality of the probabilistic forecasts.

A. Day-Ahead Forecasting Results Across Seasons

Table I presents the day-ahead forecasting performance of the three models evaluated across four seasons—winter, spring, summer, and fall. The performance is measured using NRMSE, MAPE, and CRPS. Fig. 2 illustrates the forecasted load profile and uncertainty bands for a representative spring day using the MC Gaussian LSTM model. Table I shows that the deterministic LSTM achieves the lowest NRMSE and MAPE across nearly all seasons, confirming its strength in point forecasting. However, it does not offer any uncertainty estimation, which is critical for operational planning. The Gaussian LSTM provides meaningful probabilistic forecasts with competitive CRPS values, especially in winter and spring. The MC Gaussian LSTM demonstrates slightly better calibration in the summer and fall, as indicated by its lower CRPS values during those periods. Fig. 2 shows the prediction intervals generated by the MC Gaussian LSTM on a representative spring day. The purple line represents the forecasted mean load, while the shaded red bands illustrate different confidence intervals. The actual load (blue line) generally falls well within the central bands, indicating the model's reliability in capturing temporal load variability.

TABLE II
WEEK-AHEAD FORECASTING PERFORMANCE ACROSS SEASONS

Model	Metric	Winter	Spring	Summer	Fall
LSTM	NRMSE (%)	4.25	5.46	4.68	3.77
	MAPE (%)	3.25	4.53	3.56	3.02
Gaussian LSTM	NRMSE (%)	3.50	5.40	6.70	5.00
	MAPE (%)	2.80	4.30	4.80	4.20
	CRPS (MW)	323.19	422.21	548.25	423.95
MC Gaussian LSTM	NRMSE (%)	5.20	4.80	10.30	6.00
	MAPE (%)	4.40	3.70	6.80	6.00
	CRPS (MW)	450.30	373.84	1004.44	579.98

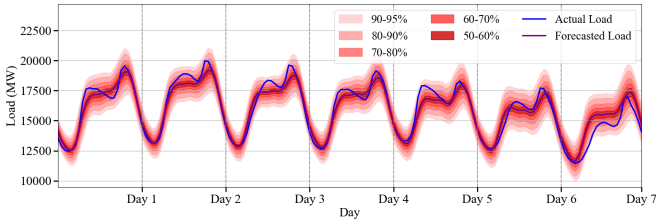

Fig. 3. Week-ahead probabilistic forecasts using MC Gaussian LSTM.

B. Week-Ahead Forecasting Results Across Seasons

Table II summarizes the week-ahead forecasting performance for winter, spring, summer, and fall using the three LSTM-based models. The deterministic LSTM is evaluated using NRMSE and MAPE, while Gaussian and MC Gaussian LSTM models also include CRPS to reflect the quality of their predictive distributions. Fig. 3 visualizes the seven-day forecast from the MC Gaussian LSTM model on a representative seasonal week. As shown in Table II, the deterministic LSTM performed well in terms of point forecasting, especially in summer and fall, achieving the lowest NRMSE and MAPE in those seasons. The Gaussian LSTM displayed a more consistent trade-off between accuracy and distribution quality, as seen in its balanced MAPE and CRPS across all seasons. The MC Gaussian LSTM, while slightly less accurate in terms of deterministic metrics, provided the broadest coverage in uncertainty, particularly evident in the high CRPS values during summer. Fig. 3 illustrates the MC Gaussian LSTM's

TABLE III
FORECASTING PERFORMANCE DURING MAJOR HOLIDAYS: JULY 4TH AND CHRISTMAS WEEK

Model	Metric	July 4th (day-ahead)	Dec. 25–31 (week-ahead)
LSTM	NRMSE (%)	4.43	6.68
	MAPE (%)	4.05	5.60
Gaussian LSTM	NRMSE (%)	2.31	6.80
	MAPE (%)	1.90	5.70
	CRPS (MW)	238.14	561.02
MC Gaussian LSTM	NRMSE (%)	3.60	6.61
	MAPE (%)	2.65	5.43
	CRPS (MW)	267.54	550.23

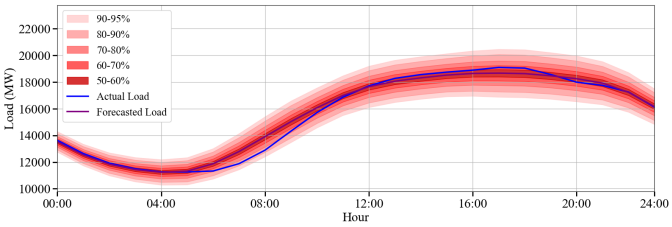

Fig. 4. Day-ahead probabilistic load forecasts using MC Gaussian LSTM for July 4th, illustrating different PIs and actual load behavior.

predicted load range over a full week, showing the forecast mean alongside varying confidence intervals. The wider spread in the early and late parts of each day reflects both temporal demand fluctuations and growing uncertainty over time.

C. Holiday Forecasting Evaluation

Holidays often exhibit atypical electricity consumption patterns due to behavioral shifts in residential and commercial sectors. Table III presents forecasting results for the Fourth of July holiday in the U.S.A. The NRMSE (2.31%) and MAPE (1.90%) values indicate that the Gaussian LSTM significantly outperformed the deterministic LSTM model (with NRMSE 4.43% and MAPE 4.05%) for day-ahead horizon, while also achieving lower CRPS (238.14 MW). This indicates that it is better at capturing the uncertainty and variability inherent to holiday load patterns. MC Gaussian LSTM model also demonstrated robust performance in day-ahead forecasting, although slightly less accurate than the Gaussian LSTM in deterministic metrics as shown in Fig. 4. Week-ahead results show that Gaussian LSTM maintained competitive performance across both accuracy and uncertainty metrics, while MC Gaussian LSTM displayed marginally better performance, highlighting the strengths of this approach.

D. Forecasting Under Extreme Demand Scenarios

As shown in Table IV, the Gaussian and MC Gaussian LSTM models demonstrate competitive performance in handling extreme demand cases. For day-ahead forecasts, the deterministic LSTM model achieved the lowest forecast error (2.02%) on the highest (10,473 MW) demand day, whereas the probabilistic models (Gaussian LSTM and MC Gaussian LSTM), though maintaining competitive deterministic forecasts, lagged behind. The Gaussian LSTM produced higher errors in general, with a marginally lower MAPE (3.03%) on the lowest (22,960 MW) demand day. The MC Gaussian LSTM

TABLE IV
DAY-AHEAD AND WEEK-AHEAD LOAD FORECASTING DURING LOWEST AND HIGHEST DEMAND CONDITIONS

Model	Metric	Day-ahead		Week-ahead	
		Low	High	Low	High
LSTM	NRMSE (%)	4.05	2.44	6.54	3.51
	MAPE (%)	3.07	2.02	4.85	2.61
Gaussian LSTM	NRMSE (%)	4.39	3.51	8.90	10.47
	MAPE (%)	3.03	2.97	7.51	6.90
	CRPS (MW)	345.91	314.88	650.11	964.27
MC Gaussian LSTM	NRMSE (%)	5.70	3.60	7.04	9.83
	MAPE (%)	4.80	2.70	5.89	6.63
	CRPS (MW)	436.80	309.67	500.51	939.49

TABLE V
PICP (%) FOR WEEK-AHEAD AND DAY-AHEAD FORECASTS AT MULTIPLE PREDICTION INTERVALS (PIS)

Model	Scenario	50%	60%	70%	80%	90%
Week-ahead						
Gaussian LSTM	Winter	72.1	81.5	91.1	95.2	98.8
	Spring	61.2	70.6	79.5	90.2	96.2
	Summer	52.4	62.5	70.2	85.7	91.6
	Fall	57.7	66.6	79.9	91.7	97.5
	High Demand	48.8	56.5	66.7	77.4	91.7
	Low Demand	29.2	42.7	71.6	77.4	90.2
	Holiday	38.1	50.6	60.7	71.4	86.2
MC Gaussian LSTM	Winter	71.4	83.2	92.9	96.4	99.0
	Spring	64.9	73.8	92.3	94.2	97.5
	Summer	64.9	73.8	82.7	91.6	97.5
	Fall	74.4	82.2	93.5	98.0	99.0
	High Demand	55.4	66.7	76.2	91.7	91.7
	Low Demand	41.4	50.9	63.0	82.6	87.5
	Holiday	30.4	39.3	51.2	67.9	80.4
Day-ahead						
Gaussian LSTM	Winter	75.0	83.3	87.5	100.0	100.0
	Spring	66.7	75.0	83.3	95.8	100.0
	Summer	50.0	66.7	83.3	95.8	100.0
	Fall	83.3	91.7	100.0	100.0	100.0
	High Demand	83.3	91.7	100.0	100.0	100.0
	Low Demand	58.3	58.3	66.7	75.0	100.0
	Holiday	100.0	100.0	100.0	100.0	100.0
MC Gaussian LSTM	Winter	58.3	58.3	79.2	91.7	91.7
	Spring	75.0	87.5	91.7	95.8	100.0
	Summer	58.3	66.7	79.2	100.0	100.0
	Fall	100.0	100.0	100.0	100.0	100.0
	High Demand	58.3	66.7	91.7	91.7	100.0
	Low Demand	20.8	20.8	33.3	62.5	95.8
	Holiday	62.5	66.7	70.9	79.2	87.5

generated better deterministic forecasts than the Gaussian LSTM, specifically in the week-ahead horizon, highlighted by a lower NRMSE and MAPE. The MC Gaussian LSTM also generated better calibrated confidence intervals as indicated by lower CRPS scores. Under week-ahead conditions, all models see a general increase in forecasting error, as expected due to the longer prediction horizon. While Gaussian LSTM shows strong performance in terms of deterministic accuracy, the MC Gaussian LSTM maintains tighter probabilistic bounds, especially under the lowest demand scenario. These results confirm that incorporating uncertainty enhances forecasting quality during both low-load and peak-stress conditions.

E. Prediction Interval Coverage Probability Analysis

To further evaluate the quality of probabilistic forecasts generated by the Gaussian and MC Gaussian LSTM models, we assess their PICP that measures the percentage of true values that fall within a predicted interval range, providing insights into the reliability and calibration of forecast uncertainty.

Table V presents the PICP results at multiple confidence levels (50% to 90%) for both day-ahead and week-ahead horizons. Both probabilistic models show strong interval calibra-

tion for most cases. The MC Gaussian LSTM provides tighter and better-calibrated intervals in several cases, particularly in fall and winter across both horizons. Notably, during low demand and holiday scenarios, coverage drops significantly, revealing the challenge of uncertainty estimation during irregular and sparse demand behavior. Day-ahead forecasts show particularly high coverage, often exceeding 90% for 90% PI, suggesting that the models are well-calibrated under normal and high-demand conditions. The week-ahead forecast results are more dispersed, with lower coverage for extreme cases and tighter alignment in seasonal scenarios, reflecting the expected uncertainty growth with forecast horizon.

IV. Conclusion

This paper presented a comparison between deterministic and probabilistic deep learning-based load forecasting models using LSTM architectures. A baseline deterministic LSTM was developed for more accurate point forecasting, while two probabilistic models (Gaussian LSTM and MC Gaussian LSTM) were developed to capture uncertainty associated with deterministic forecast model. The models were trained on historical hourly electric load and temperature data, and their performance was evaluated for day-ahead and week-ahead forecast horizon, and under diverse scenarios such as seasonal variation, holidays, and extreme demand conditions.

Test results demonstrated that the deterministic LSTM model consistently achieved a higher degree of forecast accuracy, however, it lacks the capability to quantify uncertainty that is crucial for risk-aware operational decision-making. The Gaussian LSTM model offered a balanced trade-off between accuracy and probabilistic calibration, thereby delivering reliable prediction intervals with competitive CRPS scores across most scenarios. The MC Gaussian LSTM model, although occasionally less accurate in terms of deterministic metrics, proved to be more robust in uncertainty representation especially during high-variability periods like summer and fall. A detailed PICP analysis further validated the effectiveness of both probabilistic models with the Gaussian LSTM dominating day-ahead coverage (often 100% at the 90% PI) and outperforming the MC Gaussian LSTM in the week-ahead low-demand and holiday cases, whereas the MC Gaussian LSTM attained higher week-ahead coverage in typical seasonal weeks. These findings underscored the importance of integrating uncertainty quantification into load forecasting pipelines. Note that test results presented in this paper are data-specific to the ISO−New England for the year 2011 and may not generalize to other years or regions.

Future work would be interesting to utilize the probabilistic load forecasts information for solving unit commitment and economic load dispatch problems.

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